

**PROJECT NAME: Smart Solar– AI-Driven
Solar–Biogas System for Sustainable Communities**

TEAM: ESPADA

SRI SAI RAM ENGINEERING COLLEGE

IEEE YESIST TRACK: Innovation Challenge

Team Members:

Vigneshwaran J	— Project Lead and System Architect
Divya Dharshini A	— Web Dashboard and Software Lead
Raghav G S	— Embedded Systems and Presentation Lead
Hemamalar D	— Technical Documentation & Research Lead

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ABSTRACT

The Smart Solar–Biogas Integrated Community Cooking System is a decentralized renewable energy solution that converts household and agricultural organic waste into methane-rich biogas for clean cooking. It addresses rising LPG costs, poor waste management, and environmental pollution in rural and semi-urban communities by creating a circular waste-to-energy model.

The system uses solar-powered IoT sensors to monitor key digestion parameters, including temperature, pH, and gas pressure. An AI-based optimization framework analyzes real-time data to regulate feeding, stirring, and heating cycles, ensuring stable and efficient biogas production while reducing operational losses.

The generated gas is stored in centralized tanks or compressed into cylinders for structured distribution to 15–30 households. Surplus gas can be sold to generate community income, while a mobile application enables booking, monitoring, and transparency in usage and carbon savings.

Overall, the solution promotes environmental sustainability, reduces fossil fuel dependence, strengthens community energy security, and offers a scalable, technology-driven clean cooking model.

INTRODUCTION

Access to clean and affordable cooking energy remains a challenge in many rural and semi-urban communities. Most households rely on LPG and firewood, while large volumes of organic waste are poorly managed, leading to environmental pollution. Traditional biogas systems exist but lack smart monitoring and structured distribution, leading to inefficient and unstable gas production.

This situation results in high fuel costs and underutilized waste resources. Without real-time monitoring of parameters like temperature and pH, methane production becomes inconsistent, and surplus gas is rarely stored or distributed efficiently.

This issue is important because clean energy and sustainable waste management support SDG 7 (Affordable and Clean Energy), SDG 11 (Sustainable Cities and Communities), and SDG 13 (Climate Action).

This project proposes a Smart Solar–Biogas Integrated Community Cooking System that uses AI-based monitoring, solar-powered sensors, and app-based distribution to deliver efficient and sustainable clean cooking energy.

PROBLEM STATEMENT

Existing Systems

Most rural and community-level cooking systems rely on LPG, firewood, or conventional biogas plants. Traditional biogas units operate with minimal monitoring and manual control. They do not track real-time parameters such as temperature, pH, or gas pressure, resulting in unstable and inconsistent gas production.

Resource Inefficiency

Large amounts of household and agricultural organic waste are discarded instead of being converted into energy. At the same time, communities continue spending heavily on LPG. Poor waste utilization and inefficient digestion processes lead to lower methane yield and missed opportunities for renewable energy generation.

Lack of Smart Monitoring and Distribution

Conventional biogas systems lack intelligent control mechanisms and digital platforms. They cannot optimize feeding, stirring, or heating cycles automatically. Surplus gas is rarely stored in cylinders or managed through structured distribution systems, limiting scalability and revenue potential.

Rural Energy Challenges

Rural and semi-urban areas often face rising fuel costs and limited energy infrastructure. Despite lower and irregular demand patterns, fuel expenses remain high. The absence of smart, affordable, and scalable clean cooking solutions increases financial burden and environmental impact in these communities.

LITERATURE REVIEW

Traditional Biogas Systems

Conventional anaerobic digestion systems are widely used to convert organic waste into biogas for cooking. They provide a renewable alternative to LPG and firewood and support basic waste management. However, these systems usually operate manually, without real-time monitoring of parameters like temperature, pH, or gas pressure, leading to inconsistent methane production.

Solar-Assisted Biogas Units

Some studies have explored integrating solar energy to maintain digester temperature and power basic monitoring devices. This improves energy independence and supports stable operation. However, most systems focus only on heating support and lack intelligent automation or predictive control mechanisms.

IoT-Based Monitoring Systems

Recent research proposes IoT-enabled sensors for tracking digestion parameters and gas output. These systems enable remote monitoring and data collection. However, many rely heavily on continuous internet connectivity and do not incorporate AI-based optimization for adaptive control.

Key Limitations

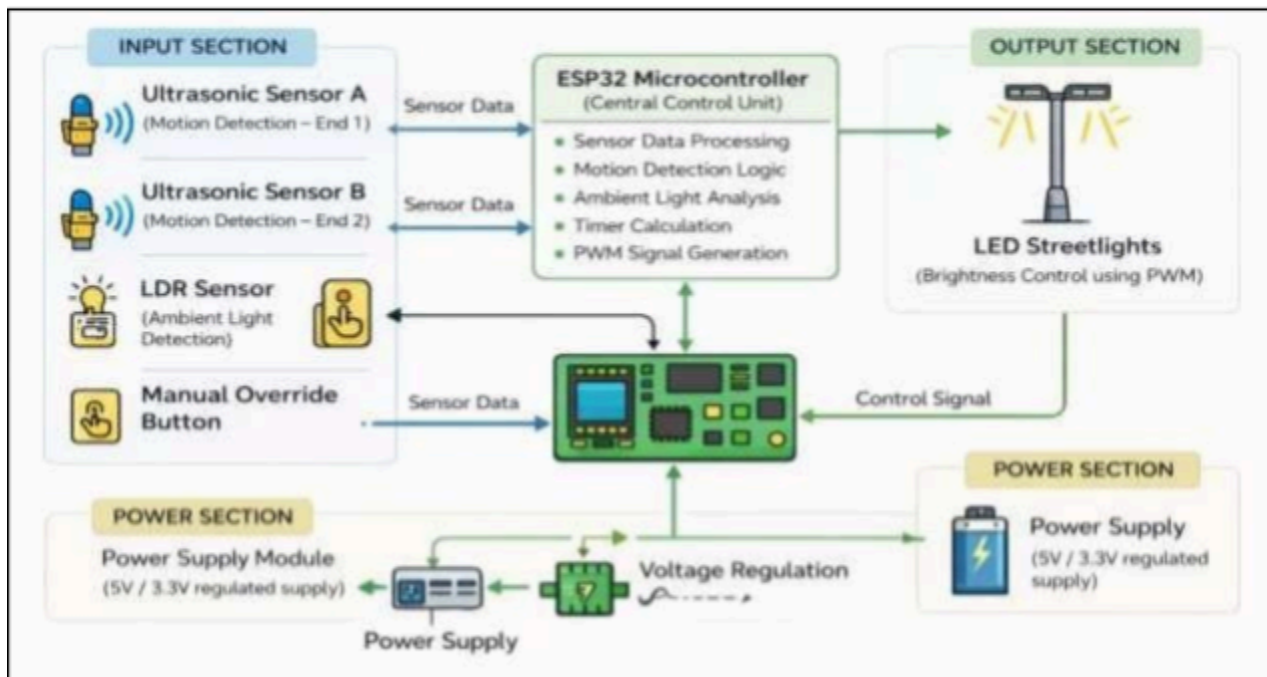
Limited real-time optimization, absence of AI-driven predictive control, high deployment cost for advanced systems, lack of structured gas storage and digital distribution models, and minimal scalability in rural community-level applications.

PROPOSED SYSTEM / METHODOLOGY

System Architecture

The proposed Smart Solar–Biogas Integrated Community Cooking System uses solar-powered IoT sensors and a microcontroller to monitor digester conditions, while an AI-based system optimizes gas production and manages structured storage and app-based distribution at the community level.

Block Diagram



Hardware Components

Component	Key Function	Component	Key Function
Anaerobic Digester Tank	Converts organic waste into methane-rich biogas through anaerobic digestion.	Gas Pressure Sensor	Tracks biogas pressure for safe storage and distribution.
Temperature Sensor	Monitors digester temperature to maintain optimal microbial activity.	Gas Flow Sensor	Measures the amount of biogas produced and supplied.
pH Sensor	Measures acidity levels to ensure stable methane production.	ESP32	Collects sensor data and controls automation processes.

Stirring Motor	Mixes the slurry to improve digestion efficiency.	Heating Element	Maintains required temperature during low ambient conditions.
Solar Panel	Provides renewable power for sensors, controllers, and automation units.	Biogas Storage Tank	Stores generated gas safely for community distribution.

Control Algorithm – Default Operation Mode

Under normal operating conditions, the system maintains optimal digestion parameters using predefined temperature and pH thresholds. If all parameters remain within the safe range, the digester operates in standard mode with scheduled stirring and feeding cycles to ensure stable biogas production.

Sensor-Based Monitoring Logic

If the temperature drops below the required range or pH deviates from optimal levels, the system automatically activates heating elements or adjusts feeding rates. If gas pressure exceeds the safety threshold, the system regulates output or triggers safety valves to prevent overpressure.

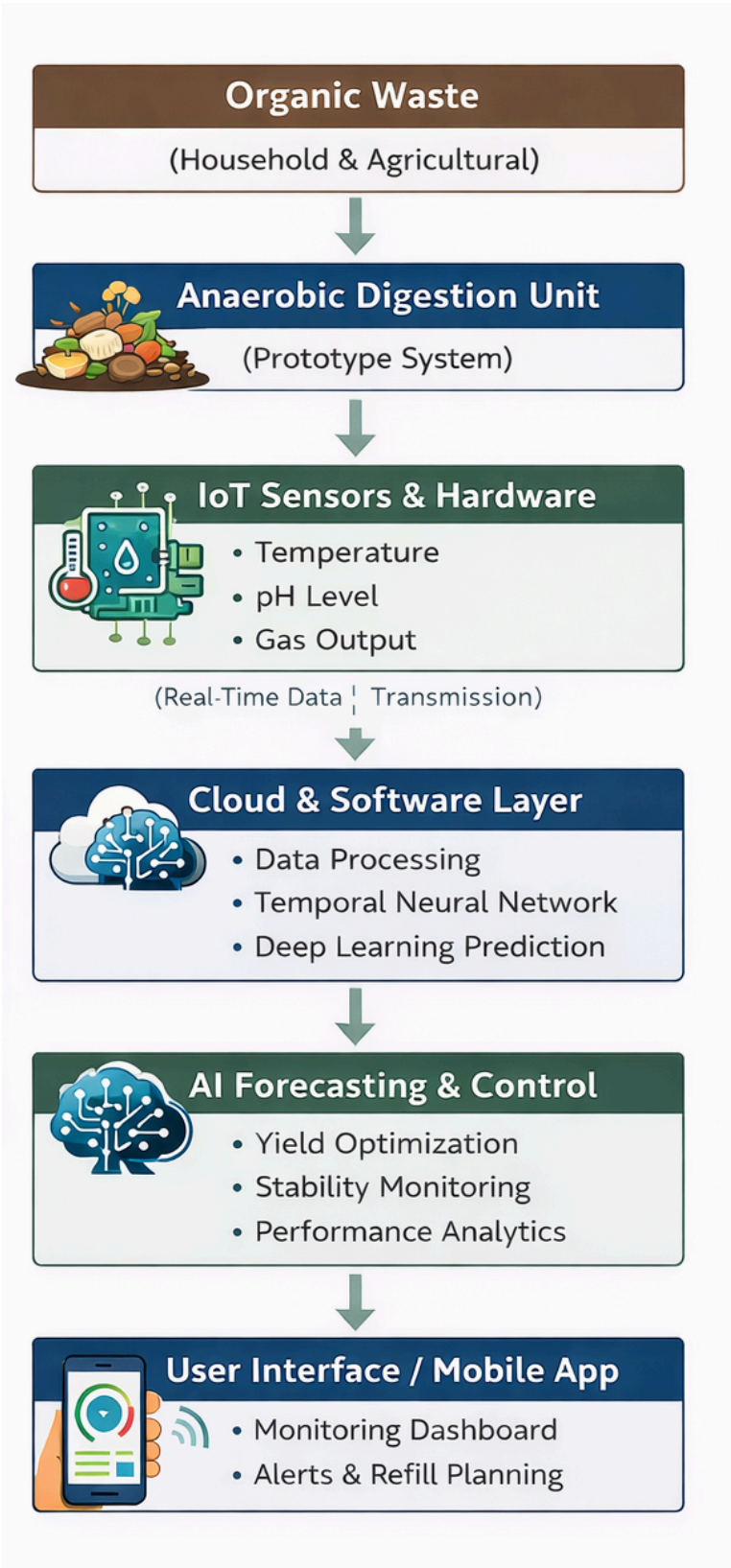
AI-Based Optimization Logic

The AI model continuously analyzes real-time and historical data to predict gas yield and adjust stirring frequency, heating duration, and feed intervals. This adaptive control ensures maximum methane efficiency and reduces operational instability.

Manual Override Logic

A manual control switch is provided to allow operators to override automated settings during maintenance or emergency conditions. After a fixed duration or once parameters stabilize, the system automatically returns to AI-controlled normal operation.

Flowchart



FEASIBILITY&SCALABILITY

Cost Estimation

Component	Qty	Estimated Cost (USD)
ESP32	1	\$4
Anaerobic Digester Tank	1	\$360
Temperature Sensor	2	\$8
pH Sensor	1	\$25
Gas Flow Sensor	1	\$25
Gas Pressure Sensor	1	\$15
Stirring Motor	1	\$30
Heating Element	1	\$30
Solar Panel	2	\$150
Biogas Storage Tank	1	\$100
Total Estimated Cost	—	\$750

Economic & Rural Feasibility

The proposed system is designed to be cost-effective and practical for deployment in rural and semi-urban areas where affordability and resource optimization are essential.

- Utilizes locally available organic waste as feedstock, reducing dependence on expensive LPG.
- Potentially reduces household cooking fuel expenses by 30–50% compared to conventional LPG usage.
- Solar-powered monitoring reduces electricity expenses and operational costs.
- Structured gas storage and surplus sales generate additional community revenue.
- Uses low-cost hardware components such as ESP32, basic sensors, and modular digesters to keep installation costs manageable.
- Does not require continuous high-speed internet connectivity, making it suitable for rural regions.
- Easily scalable for clusters of 15–30 households without major infrastructure modifications.

Scalability and Future Integration

The proposed system can be scaled from a small community prototype to a multi-village decentralized energy network with the following future enhancements:

- IoT-Based Centralized Monitoring Dashboard
- Advanced Solar Integration
- Gas Purification Modules
- AI-Based Predictive Maintenance
- Carbon Credit and Emission Tracking
- Data Analytics for Energy Optimization

CONCLUSION

The proposed system provides a renewable and eco-friendly alternative to conventional cooking fuels by converting organic waste into clean biogas. It promotes sustainable energy generation while reducing dependence on fossil fuels.

The system significantly reduces environmental impact by lowering carbon emissions and improving waste management practices. It supports the responsible utilization of renewable resources and contributes to a cleaner environment.

Economically, the system reduces household LPG expenses and creates opportunities for additional income through surplus gas or by-products. This enhances financial stability for communities.

With features such as solar-powered monitoring, AI-based optimization, and app-based management, the model is smart, efficient, and scalable for community-level implementation.

Overall, the system offers a practical, affordable, and sustainable clean energy solution that strengthens energy independence and supports long-term development.